Architecture Justification

Pharmacy Inventory Management System | Lab 3

Selected Architecture: Layered Architecture

The Pharmacy Inventory Management System uses a layered architecture that separates the presentation layer, pharmacy business operations, and persistent data. The Staff Dashboard and Supplier Portal form the presentation layer. Inventory and Batch Management, FEFO Stock Allocation, Expiry Monitoring and Alerts, Purchase Order Management, and Reports form the business layer. The Pharmacy Database stores system records.

Two Scenario-Specific Reasons

• **Consistent stock handling:** Batch records, expiry dates, and FEFO allocation are closely related. Keeping these rules in the business layer provides a clear place to validate stock availability and select the earliest-expiring eligible batch before dispensing.

• **Controlled replenishment workflow:** Expiry and stock-level monitoring can pass replenishment needs to Purchase Order Management, while supplier-facing access remains separated from internal pharmacy operations. This supports automatic purchase-order and supplier-order requirements without exposing internal inventory functions.

Security Advantage

The separate presentation and business layers allow role-based access checks to be enforced before supplier requests reach purchase-order data. Suppliers can be limited to dispatched orders relevant to them, while staff-only inventory, batch, and dispensing operations remain protected behind the business layer.

Performance Benefit

The FEFO and batch-selection logic can operate within the business layer using indexed batch and expiry-date records in the data layer. This avoids mixing user-interface work with stock-selection rules and supports efficient inventory queries and the Lab 1 performance and availability goals.